

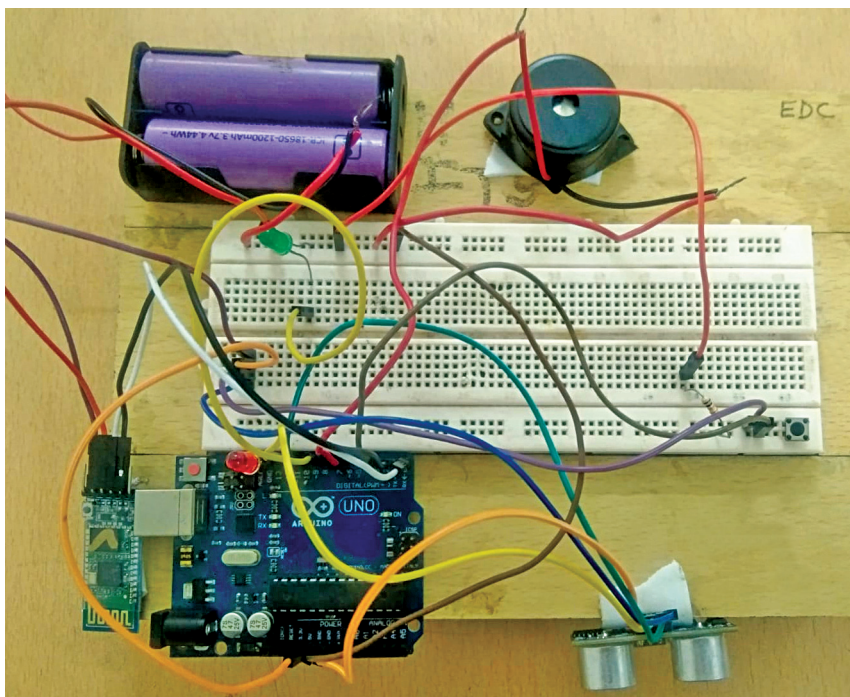


Fig. 3: Author's prototype

Arduino Uno board and the other devices used for the gadget.

The circuit is built around Arduino Uno (MOD1), ultrasonic sensor HC-SR04 (S1), HC-05 Bluetooth module (MOD2), buzzer (B1), and LEDs. The ultrasonic sensor is used to maintain the proper distancing, and a Bluetooth module is used to view the distance on mobile phones via the serial terminal/monitor of the Bluetooth. The red LED (built into Arduino Uno) and buzzer are used to alert violation of the 2-meter social distancing norm. While the green LED (connected to D8 pin of Arduino Uno) is used as the 20-second restricted time period indicator.

The working of the gadget is simple. As soon as the distance of the person using this device becomes less than two metres, the red LED glows, and the buzzer sounds. This distance can also be seen on a mobile phone via the Bluetooth monitor.

For hand-washing, when the push-to-on switch is pressed, it enables the green LED for 20 seconds. The green LED remains on for 20 seconds to help you wash hands properly with soap. Once the LED turns off, one can stop rubbing hands with soap and the hands can now be washed with water.

To build this gadget, the code is created in Arduino IDE. First of all, pins are defined for LED, buzzer, ultrasonic sensor, and then in the loop function the sensor data is checked. Functions are sent for alert and others. Then, by selecting the port, the code is uploaded. Fig. 2 shows a screenshot of the source code.

EFY Note

The source code of this project is available for free download at www.electronicsforu.com



Fig. 4: Phone showing the sensor data on a Bluetooth terminal monitor

Construction and testing

Assemble the circuit based on the circuit diagram, referring to Fig. 1 and Table 2. After proper assembling, power the device. Once it is powered, refer to the author's prototype shown in Fig. 3. Now, on a phone, install a serial terminal app and connect it with the Bluetooth HC-05 of the device.

The distance can be seen on the phone in real time. When the distance is too close, the device will alert and also show in the serial terminal in real time, as shown in Fig. 4. **EFY**

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